

Aniket Dev Roy

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EDUCATION

Purdue University
M.S. Geo-Data Sciences

West Lafayette, IN
Aug 2024 – Dec 2025

Purdue University
B.S. Earth Sciences

West Lafayette, IN
Aug 2020 – May 2024

GIS Professional Certificate (ID: 25I2-0NIR-AOYY)

EXPERIENCE

Purdue University Libraries
GIS Analyst

West Lafayette, IN
Jan 2025 – Present

- Executed a statewide food desert analysis across 1,500+ Indiana census tracts integrating Python (GeoPandas), Census TIGER/Line shapefiles, USDA Food Access Research Atlas, and ACS demographics via Census API, producing a parameterized pipeline replicable across all 50 U.S. states.
- Built network-based grocery accessibility workflows in QGIS using shortest-path routing and service-area generation across 92 Indiana counties on OpenStreetMap road networks, replacing straight-line distance assumptions with street-level access metrics.
- Authored 6 open-source Carpentries-based GIS training modules covering the full geospatial data lifecycle for an NSF-funded Purdue cybertraining project, published via GitHub Pages and actively used by 100+ researchers statewide.
- Taught hands-on QGIS, remote sensing, and Python sessions to 40+ junior and senior GIS professionals at a 2-day Purdue-NSF GIS Cybertraining workshop, covering end-to-end spatial workflows from data collection to visualization.

Purdue University – College of Science
Research Scientist

West Lafayette, IN
May 2023 – Aug 2024

- Benchmarked PCA and a custom grid-search algorithm for estimating near-surface shear-wave velocities, generating 500+ synthetic seismic datasets via Linux shell scripting across frequency, layer-thickness, and time-window parameter spaces.
- Applied both methods to broadband data from 60 seismic stations across Northern California using 16 quality-checked earthquakes, producing 4 spatial hazard maps flagging high ground-shaking risk zones across the study region.
- Co-authored an abstract at AGU Fall Meeting 2024 and delivered a first-author poster at the Purdue Undergraduate Research Symposium, presenting findings to 300+ researchers across earth and atmospheric sciences.

PROJECTS

Global Tropical Cyclone Intensity Trends | *M.S. Capstone*

Aug – Dec 2025

Python, SciPy, Cartopy, xarray, statsmodels | *NOAA IBTrACS & HURSAT NetCDF*

- Built an end-to-end Python pipeline analyzing 45 years (1980–2024) of global tropical cyclone records across 7 ocean basins, automating ingestion and preprocessing of NOAA IBTrACS and HURSAT satellite NetCDF data into analysis-ready $4^\circ \times 4^\circ$ gridded datasets.
- Applied Theil–Sen estimators and bootstrap confidence intervals across 5+ storm-strength thresholds and multiple time periods, cross-validating trends between 2 independent datasets and isolating intensification concentrated in major hurricanes (≥ 64 kt), strongest in the North Atlantic.

Ozone vs. COVID-19 Spatial Analysis | *Purdue University*

Aug – Dec 2024

ArcGIS Pro, ArcGIS StoryMaps, Python (pandas) | *3-County EPA & Public Health Data*

- Designed a multi-county spatial analysis in ArcGIS Pro correlating ground-level ozone with COVID-19 infection rates across 3 climatically distinct U.S. counties using 2022 EPA air quality and public health datasets spanning 10+ monitoring sites per region.
- Produced 8+ choropleth maps, road-network medical accessibility scores, and correlation scatterplots; published findings as an interactive ArcGIS StoryMap, surfacing regionally inconsistent correlations (0.67 LA, 0.36 Marion, ~ 0 Miami-Dade) across 3 climate zones.

TECHNICAL SKILLS

GIS Platforms: ArcGIS Pro, QGIS, ArcGIS Online, ArcGIS StoryMaps, QField, Survey123, Experience Builder

Programming: Python (GeoPandas, pandas, NumPy, xarray, SciPy, Matplotlib, Cartopy), MATLAB, Linux/Bash, Git/GitHub

Geospatial Data: Census API/ACS, TIGER/Line, OpenStreetMap, USDA FARA, NetCDF, shapefiles, network analysis, CRS management

Core: Cartography & Thematic Mapping, Reproducible Workflows, Technical Documentation, Instruction & Training